Every logarithmic Smirnov function extends to the Hilbert polydisc

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Abstract

Every radial Nevanlinna function on the ℓ^1 polydisc extends uniquely and Fréchet holomorphically to the Hilbert polydisc. The latter is exactly the domain of continuous polynomial evaluation for the logarithmic metric. The conclusion concerns exactly the catalogue target; no historical priority claim is made.

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1 Statement and argument

For $q = 1, 2$, write $D_q^\infty = \ell^q(\mathbb{C}) \cap \mathbb{D}^\infty$, with its ℓ^q topology. The space $N_*(\mathbb{T}^\infty)$ is the closure of analytic polynomials depending on finitely many coordinates for the metric

$$d_0(f, g) = \int_{\mathbb{T}^\infty} \log(1 + |f - g|)\, dm.$$

The answer to the existence question is negative: every element of $N_*(\mathbb{T}^\infty)$ has a Fréchet-holomorphic extension to D_2^∞, agreeing with its canonical representative on D_1^∞.

The key is a dimension-independent estimate for the logarithmic metric. We construct a bounded averaging density from analytic discs; the ordinary product Poisson density itself need not be bounded uniformly in the dimension. No infinite-dimensional change of variables is needed.

Write m_N for normalized Haar measure on $\mathbb{T}^N$. For $0 < r < 1$ and $M \geq 0$, set

$$C(r, M) = \exp\left(2M + \left[1 + \frac{1}{1 - r^2}\right] M^2\right). \tag{1}$$

Lemma 1.1 (Dimension-independent logarithmic evaluation)**.** *For every $N \geq 1$, every $z \in \mathbb{D}^N$ with $\max_j |z_j| \leq r$ and $\sum_j |z_j|^2 \leq M^2$, and every analytic polynomial P in N variables,*

$$\log(1 + |P(z)|) \leq C(r, M) \int_{\mathbb{T}^N} \log(1 + |P(w)|)\, dm_N(w). \tag{2}$$

Proof. Define the coordinatewise disc automorphism

$$T_z(v)_j = \frac{z_j + v_j}{1 + \overline{z_j} v_j}.$$

The pushforward $(T_z)_* m_N$ has the product Poisson density

$$P_z(w) = \prod_{j=1}^N \frac{1 - |z_j|^2}{|1 - \overline{z_j} w_j|^2}.$$

*The solutions and partial results in this document were found by AI. The supplied research and the separate repository review are AI work; human peer review and formal verification are not claimed.

In particular,

$$P_z(T_z(v)) = \prod_{j=1}^{N} \frac{|1 + \overline{z_j} v_j|^2}{1 - |z_j|^2}. \tag{3}$$

For $v \in \mathbb{T}^N$, put

$$S_z(v) = \sum_{j=1}^{N} \overline{z_j} v_j, \qquad Z_z = \int_{\mathbb{T}^N} e^{-2|S_z(v)|}\, dm_N(v), \qquad h_z(v) = \frac{e^{-2|S_z(v)|}}{Z_z}. \tag{4}$$

The density h_z is continuous and positive, and $h_z m_N$ is a probability measure. Crucially,

$$h_z(\lambda v) = h_z(v) \qquad (|\lambda| = 1). \tag{5}$$

For each fixed $v \in \mathbb{T}^N$, the holomorphic disc $\lambda \mapsto T_z(\lambda v)$ has center z and extends holomorphically past the closed unit disc. The function $\log(1 + |P(T_z(\lambda v))|)$ is subharmonic, so

$$\log(1 + |P(z)|) \le \int_{\mathbb{T}} \log(1 + |P(T_z(\lambda v))|)\, dm_1(\lambda).$$

Integrating against $h_z(v) dm_N(v)$ and using (5) gives

$$\log(1 + |P(z)|) \le \int_{\mathbb{T}^N} \log(1 + |P(T_z(v))|) h_z(v)\, dm_N(v). \tag{6}$$

After $w = T_z(v)$, the probability measure in this integral has density

$$g_z(w) = h_z(T_z^{-1}(w)) P_z(w).$$

We now bound this density independently of N. For $|u| < 1$, the elementary inequality $\log(1 + t) \le t$ gives

$$\log |1 + u|^2 = \log(1 + 2\,\mathrm{Re}\, u + |u|^2) \le 2\,\mathrm{Re}\, u + |u|^2.$$

Set

$$A_z = \|z\|_2^2 - \sum_j \log(1 - |z_j|^2).$$

Equation (3) then implies

$$P_z(T_z(v)) \le \exp(2|S_z(v)| + A_z).$$

Consequently

$$g_z(T_z(v)) = h_z(v) P_z(T_z(v)) \le \frac{e^{A_z}}{Z_z}. \tag{7}$$

Orthogonality of the coordinate functions and Cauchy–Schwarz give

$$\int_{\mathbb{T}^N} |S_z(v)|\, dm_N(v) \le \left(\sum_j |z_j|^2 \right)^{1/2}.$$

Jensen's inequality therefore yields

$$Z_z \ge \exp\left(-2 \int |S_z|\, dm_N \right) \ge e^{-2\|z\|_2}. \tag{8}$$

Finally $-\log(1 - t) \le t/(1 - r^2)$ for $0 \le t \le r^2$, so

$$\frac{e^{A_z}}{Z_z} \le \exp\left(2\|z\|_2 + \left[1 + \frac{1}{1 - r^2} \right] \|z\|_2^2 \right) \le C(r, M). \tag{9}$$

Combining (6)–(9) proves (2). □

Theorem 1.2. *Every $f \in N_*(\mathbb{T}^\infty)$ admits a unique Fréchet-holomorphic function $G : D_2^\infty \to \mathbb{C}$ whose restriction to D_1^∞ is the canonical representative of f.*

Proof. By the definition of N_*, there are analytic polynomials P_k, each depending on finitely many coordinates, such that $d_0(P_k, f) \to 0$. In particular (P_k) is d_0-Cauchy. For $0 < r < 1$ and $M < \infty$, write

$$V_{r,M} = \{\zeta \in \ell^2 : \|\zeta\|_\infty \le r,\ \|\zeta\|_2 \le M\}.$$

Apply the lemma to $P_k - P_l$ using the finite set of its coordinates. Projection onto this set does not increase either norm of ζ, and the finite-dimensional Haar integral is exactly the infinite-dimensional integral for this cylindrical polynomial. Thus

$$\sup_{\zeta \in V_{r,M}} |P_k(\zeta) - P_l(\zeta)| \le \exp\big(C(r,M) d_0(P_k, P_l)\big) - 1 \longrightarrow 0. \tag{10}$$

There is therefore a well-defined limit G on D_2^∞, with uniform convergence on every $V_{r,M}$.

The set D_2^∞ is open in ℓ^2. Indeed, every $\zeta \in D_2^\infty$ has $\|\zeta\|_\infty < 1$: its coordinates tend to zero, while each of the finitely many remaining coordinates has modulus less than one. Every such point has an open Hilbert-space neighborhood contained in some $V_{r,M}$. The P_k are continuous Fréchet-holomorphic polynomials on ℓ^2; their locally uniform limit is Fréchet holomorphic on D_2^∞.

On D_1^∞, the canonical representative is the pointwise limit of the same d_0-approximating polynomials, by the construction in [1, Lemma 2.24 and Theorem 2.27]. Thus G has the required restriction. Estimate (10) also proves independence of the polynomial approximation. Uniqueness follows from continuity and the density of D_1^∞ in D_2^∞, witnessed by finite-support truncations. □

Corollary 1.3 (Extension for the full radial Nevanlinna class)**.** *Let F be holomorphic on D_1^∞ and suppose*

$$K := \sup_{0<s<1} \int_{\mathbb{T}^\infty} \log^+ \Big|F(sw_1, s^2 w_2, s^3 w_3, \ldots)\Big|\, dm(w) < \infty. \tag{11}$$

Then F extends uniquely to a Fréchet-holomorphic function on D_2^∞. Thus the conclusion holds for the full class $N(D_1^\infty)$ defined by [1, Section 2.1, equation (2.3)], as well as for its Smirnov subclass.

Proof. For $0 < s < 1$, let $D_s z = (s^j z_j)_{j \ge 1}$ and set $F_s = F \circ D_s$. The map $D_s : \ell^2 \to \ell^1$ is bounded, so F_s is holomorphic on D_2^∞. Moreover, the weighted closed polydisc

$$K_s = \prod_{j \ge 1} s^j \overline{\mathbb{D}}$$

is compact in ℓ^1 and contained in D_1^∞. Uniform continuity of F on K_s shows that its finite-coordinate truncations approximate F_s uniformly on the closed coordinate polydisc. Each truncation is holomorphic on a neighborhood of its finite-dimensional closed polydisc, and hence is uniformly approximable there by analytic polynomials. Thus F_s belongs to the infinite polydisc algebra, as also recorded immediately before the proof of [1, Proposition 2.3].

Passing to uniform polynomial limits in (2), and using $\log(1 + x) \le \log 2 + \log^+ x$, gives

$$\log(1 + |F_s(\zeta)|) \le C(r,M)(K + \log 2) \qquad (\zeta \in V_{r,M},\ 0 < s < 1). \tag{12}$$

The family $(F_s)_{0<s<1}$ is therefore locally uniformly bounded in the Hilbert-space topology on D_2^∞.

Fix a point $\zeta \in D_2^\infty$ and a Hilbert ball about it whose doubled ball lies in some $V_{r,M}$. Cauchy's estimate on complex lines gives a uniform bound for the operator norms of DF_s on the smaller ball. Consequently the F_s are equi-Lipschitz there. At every finite-support point a, $D_s a \to a$ in ℓ^1 as $s \uparrow 1$, so $F_s(a) \to F(a)$. Finite-support points are dense in ℓ^2; comparing two functions F_s, F_t at a nearby such point and using the common Lipschitz bound proves that $F_s(\zeta)$ is Cauchy as $s \uparrow 1$. Denote its limit by $G(\zeta)$.

The limit G is locally bounded by (12). On each compact disc in any complex line in D_2^∞, the common local Lipschitz bounds and pointwise convergence imply uniform convergence, by a finite net

argument. Hence the restriction of G to every such line is holomorphic. Local boundedness and this Gâteaux holomorphy imply Fréchet holomorphy. This argument does not require a finite-dimensional Montel theorem on a noncompact Hilbert ball.

Finally, for every $\zeta \in D_1^\infty$, dominated convergence gives $\|D_s\zeta - \zeta\|_1 \to 0$. Therefore $F_s(\zeta) \to F(\zeta)$, proving that G extends F. Density again proves uniqueness. □

The uniform bound (10) also excludes polynomials with arbitrarily small logarithmic metric but large evaluation at a fixed Hilbert-polydisc point. The averaging measure is correlated across coordinates through their common rotation; this is why unboundedness of the usual product Poisson density does not provide such a construction.

Corollary 1.4 (Sharpness of the Hilbert threshold)**.** *Among coordinate points $z \in \mathbb{D}^\infty$, polynomial evaluation $P \mapsto P(z)$ is continuous for the logarithmic metric if and only if $z \in \ell^2$.*

Proof. Sufficiency follows from (2). Conversely, let $z \in \mathbb{D}^\infty \setminus \ell^2$ and put $s_N^2 = \sum_{j=1}^N |z_j|^2$. For all sufficiently large N, define

$$Q_N(w) = \frac{\sum_{j=1}^N \overline{z_j} w_j}{s_N^2}.$$

Then $Q_N(z) = 1$, whereas orthogonality and Cauchy–Schwarz give

$$d_0(Q_N, 0) \le \int_{\mathbb{T}^N} |Q_N|\, dm_N \le \left(\int_{\mathbb{T}^N} |Q_N|^2\, dm_N \right)^{1/2} = s_N^{-1} \longrightarrow 0.$$

This contradicts continuity of evaluation at such a point. □

References

[1] K. Guo, J. Ni and Q. Zhou, *Nevanlinna class, Dirichlet series and Szegő's problem*, Ann. Inst. Fourier **75** (2025), 1603–1647. https://doi.org/10.5802/aif.3696. Numbered passages refer to https://arxiv.org/abs/2201.01993v2.